KMX6

Project reference | Updated 1 November 2026 | Includes isolated benchmark results

Features

A Windows virtual audio mixer with six source channels, two hardware output mixes, and four independent virtual microphone mixes. Every source can feed any combination of A1, A2, and B1-B4.

Channel / bus	Purpose
MIC 1-4	Four independently selectable hardware microphone sources.
VIRTUAL 1-2	Two fixed stereo computer-audio sources, with their Windows cable/device names shown.
A1 / A2	Hardware mixes for headphones, speakers, interfaces, or additional output pairs.
B1-B4	Independent virtual microphone mixes that receiving programs can select.

Per-channel controls

- Input gain before EQ; 80 Hz high-pass filter; Low, Mid, and High EQ.
- Channel fader from -60 to +12 dB; microphone pan or stereo balance; mute and solo.
- Animated routing switches, stereo meters, clipping indication, and frequency-responsive meter glow.
- Microphone gate and limiter. Threshold arrows are beside the meter; right-click for detailed settings.
- Each source has a PLUGINS button opening its own eight-slot effects chain.
- Inactive channels darken and become monochromatic after about ten seconds without signal. Device selectors stay readable.

Master and app controls

- Compact header knobs control B1-B4 and A1/A2. B mixes use purple accents; A mixes use gold accents. Inner rings indicate output activity and near-clipping levels.
- Virtual 1/2 offer per-app volume popouts with app icons and Windows session-volume sliders.
- Hardware inputs share a warm violet control-deck gradient; computer inputs use a cooler lavender gradient. A gap separates each source header from its controls.

Interface and everyday operation

Appearance and scaling

- Dark charcoal-purple panels, lavender/pink accents, complementary EQ colors, consistent unboxed numeric readouts, and the KMX6 mixer-cube identity.
- Six source strips share one fixed logical canvas. Scaling applies one uniform transform so controls keep their relative positions.
- SCALE is click-only: left-click increases by 5%; right-click decreases by 5%. Range: 25%-200%. Ctrl+0 resets to 100%; Ctrl+plus/minus changes by 5%.
- Zoom uses a cached image during its short animation. Synchronous window repainting was added to address the white resize area; visual behavior across displays still needs verification.
- The main window follows the scaled canvas without scrollbars. Large scales can extend beyond the screen.

Audio health at the bottom of each panel

Readout	Meaning
LOCAL ~12.3 ms	A live local estimate: capture queue depth, an engine-block allowance, and active plugin-reported delay. Hover for details and routed B-bus queue estimates. It is not microphone-to-app latency.
U / D	U counts source queue underrun events. D combines rejected block/packet events, stale-backlog discards, and Windows capture discontinuities. Hover separates event and sample counts.
--	A measurement is unavailable. Shared hardware inputs do not expose separate per-input U/D counts.

Counters reset on the relevant source/queue restart. A queue event is not proof of an audible glitch: an idle loopback source may stop supplying packets.

Device setup and recovery

- Normal startup no longer launches elevated installation or treats unchanged cable names as a setup failure.
- The existing host/link retry behavior is supplemented by non-elevating device reconnection attempts and a host watchdog. Prolonged failures show a repair status.
- REPAIR I/O sits beside Plugin Manager. It runs setup/driver repair on demand; Windows may request administrator approval.
- All six KMX6 attachments were saved for driver-managed persistence. The setup helper merges those records while preserving unrelated USB/IP records.
- Reopening was verified with four virtual-mic links and no setup process running. Reboot, sleep/wake, and prolonged device-loss recovery remain unverified.

Audio, plugins, and practical limitations

Hardware and computer audio

Native ASIO supports installed hardware drivers. WASAPI shared, low-latency shared, and exclusive modes are available. A1 sets engine timing; A2 can use another device or an additional output pair on the main interface. Four physical microphones require enough physical inputs.

Computer-audio capture now targets one incoming packet plus about 2 ms of margin, subject to minimum queue and consumer-block requirements. A typical 10 ms packet therefore requests about 12 ms rather than the former fixed 30 ms. Reserve increases after starvation and is released during stable playback; actual latency is not fixed at that target.

VST3 effects

- 64-bit VST3 mono/stereo audio effects; up to eight slots per source, with multiple independent chains active at once. VST2, instruments, and external sidechain routing are not supported.
- Slots support plugin editors, bypass, wet/dry mix, rename, replacement, move, and removal. The enable circle lights when on and is grey when off.
- Plugin Manager is separate from the slot panel. It searches standard VST3 locations and custom folders, deduplicates modules, caches results, and supports retrying failed scans.
- Live mode is a pill switch that skips individual effects reporting more than 5 ms of delay. Several effects below that threshold can still accumulate more than 5 ms in total.
- Scanning uses a separate process with a timeout. Loaded effects run inside the mixer and can crash it. No cross-channel latency compensation is implemented.
- Dry/wet alignment supports up to two seconds of reported plugin delay. A latency change can force fully wet processing until alignment is refreshed.

Implementation and limits

The C++ mixer uses JUCE; the Go endpoint host is derived from Virtual-Cables and uses USB/IP with Windows USB audio. Virtual microphones use 48 kHz, 16-bit stereo; internal mixing uses floating-point audio. The host must run for the virtual endpoints to function. VB-CABLE and Hi-Fi Cable are supported computer-audio source alternatives.

The name history is VibeMixer, OpenMix, then KMX6: K = KUI, M = Mixer, X = advanced mixing, 6 = six sources. Current credits distinguish bundled components from separately installed drivers.

Still unconfirmed: true end-to-end latency, minimum stable live buffer, real plugin-chain performance under load, long-session reliability, and behavior across all display/device configurations. A local buffer or queue reading does not cover the complete Windows and receiving-app path.

Benchmark results | 1 November 2026

Finding: the built-in six-channel mixer has substantial DSP headroom on this PC. This is an isolated KMX6 benchmark, not a measured comparison with another mixer or a measurement of end-to-end microphone latency.

Machine and workload

AMD Ryzen 5 3500X, 6 cores / 6 logical processors; approximately 16 GB RAM; Windows 11 Pro 23H2 (10.0.22631). The existing live mixer stayed at 44.1 kHz / 512 frames with its routing unchanged; saved external effects were bypassed.

The benchmark invoked the actual production callback offline with six synthetic mono sources. HPF, three-band EQ, gates, limiters, channel/bus glow analysis, six route mixes, and four output-queue writes were enabled. Output-queue consumption was outside callback timing. No external plugins or Windows audio devices were opened by the benchmark.

36 cases; 1,037,904 timed callbacks. The standard sweep covered 44.1/48/96 kHz and 64/128/256/512/1024 frames, at least one second per case. The longer 48 kHz run used five seconds baseline and ten seconds contention per setting. Its four load workers requested a nominal 75% busy duty cycle; this was bounded contention, not maximum-system-load testing.

Longer-run callback timings at 48 kHz

Scenario	Frames	Mean ms	P99 ms	Worst ms	Over budget
Baseline	128	0.042	0.101	2.077	0 / 107,906
Baseline	256	0.081	0.170	3.712	0 / 55,465
Baseline	512	0.176	0.331	65.257	6 / 25,918
Contention	128	0.043	0.121	4.986	3 / 207,730
Contention	256	0.086	0.216	31.479	7 / 105,696
Contention	512	0.169	0.387	6.513	0 / 53,573

Time budgets at 48 kHz: 128 frames = 2.667 ms; 256 = 5.333 ms; 512 = 10.667 ms. P99 means 99% of measured calls finished within that time.

At 48 kHz / 256 frames under contention, the mean callback cost was **0.086 ms**, about **1.60%** of one buffer allowance. The 99th percentile was **0.216 ms**, with **7 of 105,696** calls over budget and a worst observed call of **31.479 ms**.

Interpretation: offline timings include normal-thread scheduling preemption and do not reproduce real hardware callback scheduling. Over-budget calls show risk; they are not measured audible dropouts. Low callback cost is processing headroom, not microphone-to-app latency. These short runs are not an hours-long stability test.

Benchmark observations and comparison limits

Passive observation of the running mixer

Observation window	Mixer mean CPU	Host mean CPU	Mixer peak working set
Initial normal-use baseline	0.47%	0.02%	118.7 MiB
During longer benchmark process	0.39%	0.10%	100.8 MiB

CPU percentages are normalized to the whole six-processor PC and sampled about once per second. These are the existing live configuration, not six live hardware inputs or a large VST chain. Working-set pages may be shared. The longer window includes baseline and contention cases.

Capture delivery needs investigation

- Virtual 1/source 5 recorded nine additional underrun events between 17:11:03 and 17:12:03, then seven between 17:14:44 and 17:15:44 (Asia/Manila).
- These events existed before contention. Main-callback late/overrun counters and A2/B1-B4 underrun counters stayed at zero during both captured windows.
- No continuous known live signal was injected. Idle Windows loopback delivery can also starve a queue, so these findings do not establish audible dropouts or identify the cause.
- Final verification found one running mixer and four established virtual-mic links. Temporary benchmark processes/workers exited; no routing settings were changed and no audio recording was retained.

Other virtual mixers were not benchmarked

You selected isolated KMX6 tests only. Voicemeeter setup/removal files were found, but no runnable mixer in the inspected locations. Sonar and Wave Link were not tested. No other mixer was installed or configured. **No relative latency, CPU, or stability ranking is supported by these results.**

Voicemeeter's official latency guidance discusses driver and buffer selection; its configuration examples are not same-PC benchmark results: voicemeeter.com/how-to-optimize-your-latency-in-voicemeeter/

Recommended follow-up

- Use a continuous known signal to distinguish idle capture behavior from real missing samples before tuning queue reserves again.
- Measure render-to-receiving-client delay with signal correlation. Full microphone-input latency needs a suitable hardware or acoustic reference path.
- Run real plugin-chain, CPU/GPU/UI-contention, reconnect, sleep/wake, and longer soak tests. A fair competitor test would require matched hardware, driver, sample rate, routing, workload, and receiving client.

Files, launch reference, and validation

Current project and launch paths

Project folder: **D:\VIBE PROJECTS\KMX6**
Launch: **D:\VIBE PROJECTS\KMX6\Launch.cmd**
Or run: **D:\VIBE PROJECTS\KMX6\bin\KMX6.exe**

File / folder, relative to the project	Contents
Source/Main.cpp	UI, engine integration, routing, settings, app entry and self-checks.
Source/AudioCore.h	DSP, queues, adaptive reserves, event counters and metering.
Source/NativeCapture.cpp	Windows capture and fixed computer-source selection.
Source/PluginHost.h / PluginUI.h / PluginCatalog.h	VST3 chains, editors, scanning and plugin library.
Source/UIScale.h	Fixed-canvas scaling, controls, animation and geometry checks.
Source/MixerBenchmark.h	Isolated production-callback benchmark and contention workload.
Source/VirtualOutput.h / CableHost/	Virtual microphone transport and Go endpoint host.
Source/KMX6DeviceSetup.cpp	Setup helper that preserves unrelated saved USB/IP records.
Setup-VirtualIO.ps1	Manual setup/repair and non-elevating reconnect mode.
CMakeLists.txt / build.ps1 / Build-CableHost.ps1	Build configuration and C++ / host build scripts.

The host is **bin/KMX6CableHost.exe**. Mixer settings, UI scale, plugin state/library, audio-health logs and setup/reconnect logs are stored beside the app. The former OpenMix folder should not be used for current launches.

Recorded validation

105 regression checks passed: 55 DSP/queue, 19 plugin-feature, 16 virtual-output, and 15 scaling checks. These are separate from the 36 benchmark cases and do not certify end-to-end latency or long-session stability.

Benchmark artifacts and repeatability

The full report, standard and longer-run CSV/JSON results, timestamped process samples, health-log snapshots, regression reports, and source/build SHA-256 manifest are in:

C:\Users\charl\Documents\ChatGPT\Virtual Mixer\outputs\benchmark-2026-11-01

The benchmark-enabled build supports **--benchmark** and **--benchmark-heavy**. That build is in the project build output; the installed GUI executable was left untouched during benchmarking. No external VST performance result or true receiving-app latency result is included.

This reference updates the 31 October snapshot. It records implemented behavior and measured observations; proposed QoL ideas are not listed as existing features.